

DM-2026-002

SendCUIEmail Development Team
Software Engineering

DATE: January 21, 2026

TO: Project Record

SUBJECT: Maximum File Size Limit for Email Attachments

OFFICE OF PRIMARY RESPONSIBILITY: SendCUIEmail Project

1. **Purpose**

This memorandum establishes the maximum file size limit for SendCUIEmail encrypted attachments and documents the rationale for this limit.

2. **Decision**

Maximum supported attachment size: 10 MB (10,485,760 bytes)

Files larger than 10 MB should not be sent via SendCUIEmail. For larger files, users should use alternative secure transfer methods.

3. **Rationale**

Email Infrastructure Limitations:

- Most enterprise email systems impose attachment size limits between 10-25 MB
- Base64 encoding (used in email transmission) increases file size by approximately 33%
- A 10 MB encrypted file becomes approximately 13.3 MB after Base64 encoding
- Many email gateways reject messages larger than 10-15 MB total

Practical Considerations:

- Email is not designed for large file transfer
- Large attachments increase risk of delivery failure
- Recipients may have mailbox quotas that prevent receipt
- Email filters may quarantine or reject large attachments

Security Considerations:

- Larger files require more time to encrypt/decrypt, increasing exposure risk
- Failed transfers may result in incomplete data on intermediate systems
- Log files and email archives may retain copies indefinitely

4. Alternative Methods for Large Files

For files exceeding 10 MB, use one of these approved methods:

- **Secure File Transfer Services:** Organization-approved SFTP or cloud storage with encryption at rest
- **Encrypted USB Media:** Physical transfer using FIPS 140-2 validated encrypted drives
- **Secure File Sharing Platforms:** DoD SAFE, MilSuite, or similar approved platforms
- **Split Archives:** For files slightly over the limit, consider splitting into multiple encrypted archives (not recommended for CUI due to complexity)

5. Implementation

The file size limit is implemented as follows:

- **Warning:** Files approaching 10 MB will display a warning
- **Test Coverage:** The test suite includes a 10 MB file to verify maximum size handling
- **Documentation:** User documentation clearly states the limit
- **No Hard Block:** The tool does not prevent encryption of larger files, but users are warned

Note: This is a “moral limit” rather than a technical enforcement. Users are trusted to use appropriate methods for large file transfer.

6. Test Data

The following test files are provided in the `testdata/` directory:

File	Size	Purpose
small_text.txt	138 B	Basic text file
empty.txt	0 B	Edge case: empty file
binary_sample.bin	1 KB	Binary with null bytes
README.pdf	175 KB	Generated from README.md
medium_file.bin	1 MB	Medium binary file
large_file_10mb.bin	10 MB	Maximum size test

7. References

- RFC 5321 (SMTP): Recommends message size limits
- NIST SP 800-171: CUI handling requirements
- DoD Instruction 8170.01: Online Information Management
- Organization email policy (varies by deployment)